Image downloads: what is tried, in what order, and the fallbacks

Solar_data_mining | Multiwb_datamine.get_SDO_imgs / get_LASCO_imgs / get_STEREO_imgs

Step | Decision | Result saved / returned | Fallback or gap (logged, no crash) | Helioviewer JP2

SDO: get_SDO_imgs (AIA, HMI)

1. Jobs
One job per (instrument, product), e.g. (AIA,193), (HMI,magnetogram). nproc=1 serial, else a multiprocess Pool across jobs.

2. Search + download (find_coobs)
Fido/VSO (JSOC) search at a.Sample(dT); HMI filtered by product. Files go to outdir/<INSTR>/<wave>/raw/. Failed downloads are retried up to 3 times (10 s, 20 s back-off).

3. Prepare
AIA: Level 1 -> aiapy update_pointing + register -> Level 1.5 (0.6"/px, DN/s, north-up, PC = identity), file rewritten as *lev15*.
HMI: kept as delivered (pixels as observed, CROTA2 ~ 180 deg; header is truthful).

on error

If search/download fails
Warning; the job returns what it has (possibly an empty list). Nothing stops the other jobs.

4. Select per slot (_series)
All FITS in the product folder; the image nearest each cadence slot sT, sT+dT, ... eT is kept (duplicates dropped).

5. Save
_save_maps -> outdir/<INSTR>/<wave>/<INSTR>_<wave>_<time>.fits (full WCS, unchanged roll). PNG quicklooks.

6. Helioviewer back-up (helioviewer=True)
A JP2 for **every** cadence slot -> raw/helioviewer/ (hv<id>_<time>.jp2). If the nearest JP2 falls outside the window, one retry nudged inward. JP2s are not returned.

Return
List of FITS Map lists, one per job, in input order.

SOHO: get_LASCO_imgs (C2, C3)

1. need_calib = True? Route: Level-1 -> Helioviewer -> Level-0.5
Yes: VSO/SDAC search with a.Level(1); nearest record per slot; kept only if LEVEL >= 1 and BUNIT = MSB. Kept files saved **unchanged** in maps/ (NRL Level-1 is already north-up).

none usable

2. No usable Level-1 -> Helioviewer
A JP2 per cadence slot (cache reused; one inward retry at the window edges) -> FITS in **maps_HV/**: canonical tags (OV_SRC Helioviewer JP2, BUNIT ct), roll 0 (old roll kept in OV_ROLL), SOHO observer, occulter masked.

fails

3. Helioviewer also fails -> last fallback
Archive Level-0.5 route below (DN/s), with a warning.

Archive route (need_calib=False, or last fallback)
VSO search padded +/-30 min; if VSO fails, the LASCO team archive index (img_hdr.txt). Only standard 1024x1024 frames (C2 orange / C3 clear). Nearest record per slot; a failing file is replaced by the next nearest (up to 4 rounds). need_calib=False with helioviewer=True also saves a JP2 for every slot.

4. Prepare (_prepare_lasco)
Divide by EXPTIME -> DN/s; mask occulter and outside the FOV (C2 2.4-6, C3 3.8-30 Rsun). Pixels as observed; the spacecraft roll in CROTA (date-dependent) is kept; SOHO observer added.

5. Save + return
Level-1 and Level-0.5 in maps/, Helioviewer in maps_HV/; PNG previews. Returns the Maps.

STEREO: get_STEREO_imgs (EUVI, COR1, COR2, HI1)

1. Jobs
One job per (instrument, wave): EUVI x waves, COR1, COR2, HI1. Pool across jobs if nproc>1.

HI1 jobs only: own route (_hi_series)
Bins of max(dT, 40 min). Local Level-1 files (e.g. UKSSDC hi_prep, placed under STEREO_<sc>/HI1/) are used if they cover every bin. Helioviewer has no HI source. Otherwise SSC Level-0.5 science sums (N_IMAGES > 1, >= 512 px) through steps 2-4.

2. Reuse local FITS (_stereo_series)
Read FITS already in STEREO_<sc>/<instr>[/<wave>]/. Filters: spacecraft, detector, EUVI wavelength, COR1 polarizer = polar, **COR2 POLAR = 1001** (total brightness, d*c2 files). A slot is covered if a match lies within dT/2. SECCHI Level-1 is not distributed through VSO, so consumers use the route Helioviewer -> Level-0.5.

3. Uncovered slots only
One Fido/VSO search (SECCHI, provider SSC data archive). Per slot, download candidates nearest-first (within dT/2) until one passes the filters; failed downloads retried up to 3 times. Single-polarizer n*c2 files are rejected.

none

No match for a slot
Logged "Missing STEREO_<sc>/<instr>: time" (no crash).

4. Save
maps/STEREO_<sc>_<instr>[_wave]_<time>.fits **unchanged**: Level-0.5 secchi_rectify, DN, spacecraft roll in the PC matrix, observer HGLN/HGLT/DSUN kept; RSUN_REF written if missing. No secchi_prep, no exposure normalisation. retrieval_log.txt.

5. Helioviewer back-up (helioviewer=True)
A JP2 for every slot -> raw/helioviewer/ (EUVI, COR1, COR2; header roll 0, pixels north-up, observer kept).

Return
One Map list per job. Browse JPGs (stereo-ssc .../browse/) are never used.

After download: one product family per analysis

Route everywhere: Level-1 -> Helioviewer -> Level-0.5 (LASCO when need_calib=True; STEREO always). Science FITS live in **maps/**, Helioviewer FITS in **maps_HV/**, JP2s in **raw/helioviewer*/**. Every JP2-to-FITS conversion writes identical tags, so all Helioviewer FITS form one family. Analysis code should use the first family in the route that covers every cadence slot, otherwise the one covering most slots, and never mix families in a difference. **get_helioviewer_jp2** and **jp2_to_fits** are public.

Spacecraft roll: how orientation is set for every source

Rule: every saved FITS header truthfully describes its pixels; every consumer loads through load_map and only follows the WCS. Nothing rotates an image because of its source or level.

What each product looks like

Product	Pixels	Header roll	What happens on load (load_map)
LASCO Level-0.5 / DN/s (archive)	as observed	CROTA = spacecraft roll (date-dependent)	unchanged; SunPy applies the roll
LASCO Level-1 (NRL)	north-up	matches pixels	unchanged; no further edits
AIA Level 1.5 (aiapy)	north-up	PC = identity	unchanged
HMI (JSOC)	as observed	CROTA2 ~ 180 deg	unchanged; SunPy applies it
SECCHI EUVI/COR1/COR2/HI1 Level-0.5	as observed	PC matrix = spacecraft roll (date-dependent)	unchanged; observer kept
Helioviewer JP2 (all)	north-up	LASCO: original CROTA; others: 0	SunPy reads 0 for LASCO only via a JP2-only key
maps_HV FITS (jp2_to_fits)	north-up	0 (original in OV_ROLL)	unchanged
Helioviewer-derived FITS saved without the roll fix	north-up	stale CROTA (non-zero)	**corrected to 0 in memory**

Why Helioviewer LASCO images need care

Helioviewer delivers LASCO JP2s already rotated north-up but keeps the original spacecraft CROTA in the header. SunPy ignores that CROTA only while the JP2's *helioviewer* header key exists. A FITS copy loses the key, so if it kept the CROTA, loading it would rotate the already north-up pixels a second time by the spacecraft roll, and structures would no longer line up with sky-placed overlays such as PFSS. JP2-to-FITS conversion therefore writes roll 0 and stores the original value in OV_ROLL.

Where the roll is set in the code

- **_hv_north_up(meta)**: CROTA/CROTA1/CROTA2 -> 0, PC -> identity, old value -> OV_ROLL.
- **_get_lasco_source** (Helioviewer branch) and **jp2_to_fits** call it before saving.
- **_save_maps**: any Helioviewer-derived Map is saved with the rotation SunPy used in memory.
- **load_map**: a Helioviewer-derived image with a non-zero roll and no OV_ROLL gets roll 0 in memory; everything else is returned as it is.
- **_prepare_lasco**: verified Level-1 MSB returned untouched; otherwise DN/s + masks, roll untouched.
- Consumers rotate only via the WCS: SunPy rotate() / reproject when the rotation matrix is not identity.

Checking orientation

- A feature fixed to the instrument, such as the LASCO C3 occulter pylon, must keep the same position angle in a science FITS, its Helioviewer JP2 and any FITS made from it once each is placed by its WCS.
- The best-fit rotation between a science image and its Helioviewer counterpart, both placed by their WCS, should be about 0 deg.
- Structures in a correctly oriented image line up with overlays placed from sky coordinates (PFSS lines, limb, other instruments).

Using the images safely

- Load every image with **load_map**; never re-rotate by source or level.
- Difference only within one family: science FITS with science FITS, Helioviewer with Helioviewer.
- Helioviewer pixels are display values (0-255): fine for positions, morphology and apparent motion, not for brightness or mass.
- Headers without a reference solar radius get RSUN_REF = 695700000 m (IAU 2015 nominal) written explicitly, noted in OV_RSREF.